PYTHON PROGRAMMING



21 Days Challenge

Day-02



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Python Basics and Fundamentals - Variables, Data Types & Console Input / Output

Instructor



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Python Variables

Variables are containers for storing data values.

```
x = 10      name = ‘gobeshona’
```



Python Variables (Contd...)

- Naming Variables
- Assigning Values
- Type of Variable
- Delete a Variable
- Scope of a Variable

Naming Variables

- Start with letter or _
- Use letters, digits, _
- No start with digit
- No special characters or keywords
- Case-sensitive (Data ≠ data)

Assigning Values

- `x = 5`
- `a = b = c = 100`
- `a, b, c = 1, 1.5, 'zubaer'`

Type of Variables

type()

```
a = "5" → type(a)
b = 10 → type(b)
```

Delete a Variables

```
x = 100
del x
print(x) → Now x is undefined.
```

Scope of Variables

Local Variable

```
def func():
    x = 5
    print(x)

func()
print(x)
```

Global Variable

```
x = 10

def func():
    print(x)

func()
print(x)
```

Taking User Input

input()

```
name = input("Enter your name: ")
number = int(input("Enter your number: "))
```

Print Output

```
print()
print("Hello!", name, " Welcome to Gobeshona Learning")
```

Data Types

- Numeric Types
- Set Types
- Text Type
- Boolean Type
- Sequence Types
- Binary Types
- Mapping Type

Numeric Data Types

```
int : x = 10
```

```
Float : pi = 3.14
```

```
Complex : z = 2 + 4j
```

Sequence Data Types

```
list : x = ["apple", "banana", "cherry"]
```

```
tuple : x = ("apple", "banana", "cherry")
```

```
range : x = range(6)
```

```
string : name = "bangladesh"
```

Boolean Data Types

```
bool : x = True
```

```
bool : x = False
```

Dictionary and Sets Data Types

```
dict : x = {"name" : "John", "age" : 36}
```

```
set : x = {“apple", "banana", "cherry"}
```

ASSIGNMENT

Create three variables (name, age, height) to store your name, age, and height then print the type of each variable using the type() function.

QUESTION SESSION

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THANK YOU

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